# NPTEL - Theory of Computation

Total Points 10 **Assignment - Week 5** Pages: 2

**Question 1** (2 points)
Let $L$ be a context-free language. Then according to the pumping lemma for context-free languages, there exists an integer $p \geq 1$ such that every string $w \in L$ of length at least $p$ can be written as $w = uvwxy$ satisfying the following conditions:

- $\mid vx \mid \geq 1$
- $\mid vwx \mid \leq p$ and

  **A.** $\forall n \geq 0, uv^n wx^n y \in L$

  B. $\exists n \geq 0, uv^n wx^n y \in L$

  C. $\forall n \geq 0, uv^n wxy^n \in L$

  D. $\exists n \geq 0, uv^n w^n x^n y \in L$

There is a typo in this question. This question will be reevaluated.

**Question 2** (2 points)
Which of the following languages is not context-free?

A. $\{\}$

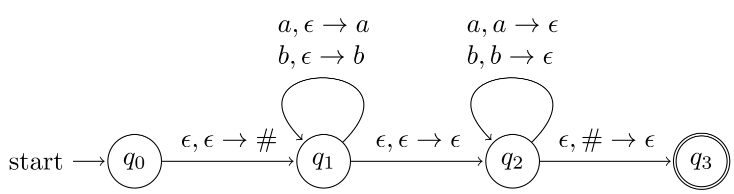
B. $\{a^n b^n w \mid m, n \geq 0, w \in \{0,1\}^*\}$

C. $\{a^n b^n c^m d^m \mid m, n \geq 0\}$

**D.** $\{ww \mid w \in \{0,1\}^*\}$

Option D is correct. Refer to Week 5 Lecture Notes (Section 10.2 Example 2)

**Question 3** (2 points)
Consider the following PDA.




What is the language accepted by this PDA?

A. Set of all strings of the form $a^n b^n$

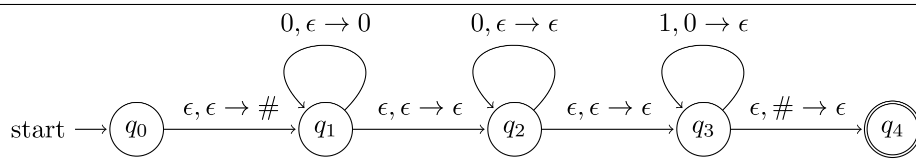
B. Set of all strings of the form $b^n a^n$

**C. Set of all even-length palindromes over the alphabet** $\{0,1\}$

D. Set of all odd-length palindromes over the alphabet $\{0,1\}$

Option C is correct. Compare the given PDA with the one in Example 2 in Section 11.1.5 of Week 5 Lecture Notes. The only difference is that the transitions $a, \epsilon \longrightarrow \epsilon$ and $b, \epsilon \longrightarrow \epsilon$ from state $q_1$ to $q_2$ are missing in the given PDA. This means the middle of the string can now only be $\epsilon$ and not $a$ or $b$. In other words, there cannot be a single middle element in the accepted string. The accepted string must be divisible into exactly two halves of equal length such that the two halves are reverse of each other. Hence, this machine accepts only even length palindromes.

**Question 4** (2 points)
Consider the following PDA.




Which of the following strings is not accepted by this PDA?

A. 00000000

B. 00000011

C. 00001111

**D. 00111111**

Option D is correct. The given PDA accepts strings of the form $0^m1^n$ such that $m \geq n$.

**Question 5** (2 points)

Which of the following is context-free?

A. $\{0^n \mid n \text{ is prime } \}$

**B. $\{ww^R \mid w \in \{0,1\}^*$ and $w^R$ is the reverse of $w\}$**

C. $\{a^nc^mb^nd^m \mid m,n \geq 0\}$

D. $\{a^nb^nc^n \mid n \geq 0\}$

Option B is correct. This is the set of all even length palindromes, which is known to be a context-free language. None of the other options are context-free languages. Refer to exercise 20 in section 10.2 of Week 5 Lecture Notes. We can show that these languages don't satisfy the pumping lemma for context-free languages.